

Hospital Admissions

Scenario Demonstration — Pack Introduction



Difficulty: 3 of 5 — Intermediate

What this pack is

This is a practice Scenario Demonstration pack modelled on the City & Guilds 9665 Data Technician end-point assessment. It contains four parts: this introduction, the Scenario Demonstration brief, the data files, and a Learner Guide showing how the task can be approached and the results you should expect to reach.

What is being looked at

This scenario exercises the full data analysis process on a larger, messier data set. The areas examined are:

Area examined	Where it appears
Importing data from multiple formats (CSV, XLSX, document table)	Scenario 1, Task 1
Validating data — spotting impossible values and missing data	Scenario 1, Task 2 (pulse, lengthofstay)
Cleaning and correcting data quality issues	Scenario 1, Task 2
Merging sources on a common key	Scenario 1, Task 3 (facid)
Deriving a classification with a formula	Scenario 1, Task 4 (BMI bands)
Aggregating and grouping (average / min / max)	Scenario 2, Tasks 1–3
Comparing a measure across categories	Scenario 2 (hospital, BMI, gender)
Exploring relationships between variables	Scenario 2, Task 4 (correlation, scatter)
Visualising and summarising findings for an audience	Scenario 2, Tasks 1–5

Why this is rated 3 stars

The tasks are clearly structured, which keeps the difficulty moderate, but the data set is large (around 10,000 records) and carries genuine data-quality traps that must be found before any analysis is trustworthy. Scenario 2 also asks for multi-factor comparison and exploration of relationships, which goes beyond a simple single-calculation task. It is not a 4-star because the analysis steps are specified rather than left open.

Areas to watch

- **pulse** contains impossible values (a negative reading and an implausibly low one) — these must be found and corrected, not left in.
- **lengthofstay** has missing (blank) records — decide how to handle them before averaging.
- Sorting or filtering a column is the quickest way to surface impossible values.
- Merge on **facid**, not hospital name — the key is the reliable join.
- Get the BMI band boundaries right: 24.9, 29.9 and 39.9 are inclusive upper limits.

What's in the pack

- **Scenario Demonstration – Hospital Admissions** (PDF) — the brief, in C&G format.
- **Admissions.csv** — primary dataset (~10,000 patient admissions).
- **Hospitals.xlsx** — hospital reference information.
- **Annex A – BMI Classification** — third source, embedded in the brief.
- **Learner Guide – Hospital Admissions** (PDF) — method and expected results.